Abstract

Public Formula 1 telemetry describes vehicle motion but does not expose battery current, voltage or the internal combustion engine-motor power split. This project develops a mathematics-guided framework for conditional energy recovery system (ERS) state estimation and constrained deployment analysis for Kimi Antonelli's 2026 telemetry. A longitudinal road-load model generates signed wheel-power demand; explicit deployment and regeneration assumptions then propagate energy continuously across laps within a modeled usable window. The initial state is imposed at 100%, and a reserve-aware policy prevents assumed deployment from consuming the configured reserve. A separate histogram-based gradient-boosting classifier identifies four contemporaneous driving states from speed, RPM, gear and derived acceleration. Three race datasets support temporal and cross-circuit classification experiments. A gradient-based constrained optimizer reallocates an Australian-lap baseline against a speed-weighted energy proxy. Temporal classification accuracy is 87.98-90.72%, while cross-track accuracy is 80.04-85.91%; these scores do not measure ERS accuracy. A stored reserve-aware optimization increases the proxy from 1.31313 to 1.52988 at essentially identical delivered energy. Continuous accounting exhibits a maximum conservation residual of 8.55 x 10^-13 MJ in the historical Australian input. The contribution is an auditable integration of energy accounting, explicit assumptions, leakage-aware ML evaluation and constrained allocation. True battery-state accuracy, race-time improvement and event-specific regulatory compliance remain unestablished.

Keywords: Formula 1; ERS; conditional state estimation; energy accounting; gradient boosting; constrained optimization; identifiability.

1 Introduction

ERS strategy couples electrical energy availability to propulsion and braking opportunities. An observer using public telemetry can inspect speed, throttle and braking, but cannot directly measure the electrical energy exchanged inside the power unit. Reconstructing a battery trace from these observations is therefore an inverse problem with missing information. A smooth, physically bounded trace can be useful for analysis while still being inconsistent with the actual hidden power split.

Prior race-car control studies formulate energy management using explicit powertrain models and resource constraints [1]-[5]. Battery estimation research combines electrical observations, state models and learned mappings [6]-[9]. This project investigates the more restricted setting in which these battery measurements are unavailable. Mathematical transparency is consequently as important as predictive performance.

The system combines three distinct tasks: conditional usable-energy propagation, telemetry-based operating-state classification, and hypothetical deployment allocation. Keeping these tasks separate prevents classification performance from being misinterpreted as battery-state validation. The report evidence cutoff is 2 November 2026. This is the inspection cutoff, not a verified assertion that every upstream session through that date has been downloaded.

1.1 Problem statement

Given irregularly sampled public telemetry and uncertain vehicle parameters, construct a continuous, bounded ERS usable-energy scenario; evaluate an ancillary driving-state classifier without adjacent-sample leakage; and explore deployment allocation under explicit energy and power constraints. Determine which conclusions follow from numerical consistency and which require independent electrical measurements.

1.2 Objectives

1. Preserve source identity, units, chronology and missing-data information during preprocessing.

2. Derive a signed longitudinal power demand and propagate conditional energy without resetting at lap boundaries.

3. Expose parameter assumptions, scenario dependence and structural non-identifiability.

4. Train and evaluate operating-state classification using chronological and cross-track partitions.

5. Solve a gradient-based constrained allocation problem and compare it with an energy-matched baseline.

6. Provide an interactive engineering dashboard and reproducible evidence exports.

1.3 Scope

The intended acquisition scope is all available Antonelli 2026 sessions at a frozen cutoff. Completed modeling and classification evidence currently covers Australian, Austrian and Italian GP race imports. The supplied archive additionally contains preprocessing material for other sessions; availability of those files is not evidence that every session was modeled or evaluated. Proprietary power-unit identification, measured battery SoC reconstruction, thermal modeling and demonstrated lap-time optimization are outside the completed scope.

2 Literature Review

The supplied 12-paper survey provides the starting bibliography. Recent racing energy-management and battery-estimation work is prioritized; the 2017 race-car controller and 2020 neural-network/filter study provide methodological foundations. The review is a selected analytical review rather than a systematic review with a documented exhaustive search.

2.1 Racing energy management

Salazar et al. [1] establish a model-based optimal-control foundation. Duhr et al. [2] couple gear selection with energy management using convex modeling, dynamic programming and Pontryagin's minimum principle in a Bahrain case study. Their formulation illustrates why RPM and gear matter jointly, but presumes a powertrain model unavailable to an external telemetry observer. Battery path constraints [3] and race-level allocation [5] motivate explicit state limits and a carefully chosen horizon. Javed and Samuel [4] compare model-based energy optimization with dynamic programming and rule-based control in simulation. These works support constrained allocation methods; their simulated performance is not a benchmark for public-data battery inference.

2.2 Battery estimation and machine learning

Neural-network/filter combinations [6], robust EKF estimation [7] and ML surveys [8] represent complementary approaches to battery state estimation. Their model and measurement requirements must be checked before transfer to F1 telemetry. Pillai et al. [7] address parameter uncertainty using additional states and innovation diagnostics, demonstrated with a battery simulator. Sun et al. [9] use real operating EV data and a CNN-BiLSTM-attention architecture with battery-pack information. Such electrical inputs are absent here. Replacing those inputs with vehicle speed does not establish an equivalent estimation problem.

2.3 The 2026 context and research gap

Regulation-oriented studies [10]-[12] explain changes in energy-management demands. Gaur [11] uses 2024 qualifying telemetry for scenario analysis of three circuits; those data are not measured 2026 battery traces. Regulatory commentary must be checked against the relevant FIA issue [13], [14]. Descriptive power-share statements in secondary literature are not imposed here as a universally mandated instantaneous engine/electric split.

The defensible gap is an auditable public-telemetry workflow that distinguishes feasible conditional energy histories from measured battery state, keeps energy continuous across laps, and evaluates a separate ML task across circuits. The work does not claim the first F1 energy optimizer or the first physics-informed battery estimator.

2.4 Literature comparison

The following compact comparison records the validation setting without importing unrelated numerical scores into this project. Where only a supplied survey or bibliographic record was accessible, detailed experimental sample counts and performance claims are deliberately omitted.

Study 1

S.No.	Author & Year	Paper Title
1	Mauro Salazar et al.; Nov 2017	Time-optimal control policy for a hybrid electric race car [1]

Objective / Problem	Methodology / Technique	Dataset / Sample
Hybrid race-car energy allocation	Optimal-control policy / Pontryagin analysis	Modeled race car; simulation

Key Findings	Limitations / Research Gap	Relevance to Our Work
Constrained allocation is tractable	Requires internal model; older architecture	Foundation for dynamic energy constraints

Study 2

S.No.	Author & Year	Paper Title
2	Pol Duhr et al.; 15 Jan 2021	Time-optimal gearshift and energy management strategies for a hybrid electric race car [2]

Objective / Problem	Methodology / Technique	Dataset / Sample
Joint gear and energy control	Convex modeling, DP and Pontryagin iteration	Bahrain model-based case study

Key Findings	Limitations / Research Gap	Relevance to Our Work
Gear and energy decisions interact	Not hidden-state reconstruction	Motivates joint RPM/gear features

Study 3

S.No.	Author & Year	Paper Title
3	Pol Duhr et al.; 29 Jun-2 Jul 2021 (conference)	Time-optimal Energy Management of the Formula 1 Power Unit with Active Battery Path Constraints [3]

Objective / Problem	Methodology / Technique	Dataset / Sample
Active battery-state limits	Optimal control with battery path constraints	Modeled F1 power unit

Key Findings	Limitations / Research Gap	Relevance to Our Work
State bounds affect the control policy	Independent telemetry-state validation absent	Motivates per-sample energy constraints

Study 4

S.No.	Author & Year	Paper Title
4	Hassan Javed et al.; 30 Aug 2022	Energy Optimal Control for Formula One Race Car [4]

Objective / Problem	Methodology / Technique	Dataset / Sample
Energy-optimal power split	GT-Suite, discrete/genetic optimization; DP reference	High-fidelity simulator

Key Findings	Limitations / Research Gap	Relevance to Our Work
Model-based control compared with baselines	Older ERS architecture; model-dependent	Supports explicit cost and baseline

Study 5

S.No.	Author & Year	Paper Title
5	Pol Duhr et al.; Jun 2023	Minimum-Race-Time Energy Allocation Strategies for the Hybrid-Electric Formula 1 Power Unit [5]

Objective / Problem	Methodology / Technique	Dataset / Sample
Race-level energy allocation	Race-time energy-management optimization	Modeled hybrid F1 power unit

Key Findings	Limitations / Research Gap	Relevance to Our Work
Extends allocation beyond an isolated lap	Requires performance and powertrain models	Clarifies limits of selected-lap replay

Study 6

S.No.	Author & Year	Paper Title
6	Seyedmehdi Hosseininasab et al.; 18 Nov-16 Dec 2020 (conference)	State-of-Charge Estimation of Lithium-ion Battery Based on a Combined Method of Neural Network and Unscented Kalman Filter [6]

Objective / Problem	Methodology / Technique	Dataset / Sample
Battery SoC estimation	Neural network plus unscented Kalman filter	Battery estimation study; exact sample count not audited

Key Findings	Limitations / Research Gap	Relevance to Our Work
Combines learned mapping and filtering	Electrical-data requirements differ	Future option with independent labels

Study 7

S.No.	Author & Year	Paper Title
7	Prarthana Pillai et al.; Published online 5 Dec 2023; journal issue Jul 2024	A Robust Extended Kalman Filtering Approach for State of Charge Estimation in Batteries [7]

Objective / Problem	Methodology / Technique	Dataset / Sample
Robust battery state estimation	Augmented robust EKF and innovation diagnostics	Battery simulator

Key Findings	Limitations / Research Gap	Relevance to Our Work
Addresses model-parameter uncertainty	Simulator validation; battery model needed	Motivates discrepancy diagnostics

Study 8

S.No.	Author & Year	Paper Title
8	Aya Haraz et al.; 28 Oct 2024	State-of-Health and State-of-Charge Estimation in Electric Vehicles Batteries: A Survey on Machine Learning Approaches [8]

Objective / Problem	Methodology / Technique	Dataset / Sample
Review ML battery SoC/SoH	Survey of ML approaches	Published studies; not a new F1 dataset

Key Findings	Limitations / Research Gap	Relevance to Our Work
Organizes learning-based approaches	Task/data heterogeneity	Guides label and input requirements

Study 9

S.No.	Author & Year	Paper Title
9	Changcheng Sun et al.; 2024	Data-driven state-of-charge estimation of a lithium-ion battery pack in electric vehicles based on real-world driving data [9]

Objective / Problem	Methodology / Technique	Dataset / Sample
Real-world battery-pack SoC	CNN-BiLSTM-attention; cell inconsistency features	Actual operating EV data

Key Findings	Limitations / Research Gap	Relevance to Our Work
Uses pack information for estimation	Battery-specific inputs unavailable here	Shows what independent validation needs

Study 10

S.No.	Author & Year	Paper Title
10	Oliwia Kropisz et al.; Online 5 Mar 2026; issue publication 2 Jul 2026	Impact of 2026 FIA technical regulations on Formula 1 power units, aerodynamics and energy management [10]

Objective / Problem	Methodology / Technique	Dataset / Sample
2026 design implications	Regulation and engineering analysis	Published technical sources

Key Findings	Limitations / Research Gap	Relevance to Our Work
Connects PU and aero changes	Not synchronized battery measurements	Context; no transferred vehicle parameters

Study 11

S.No.	Author & Year	Paper Title
11	Anushka Gaur; 29 Jun 2026	Strategic Implications of Driver-Controlled Energy Deployment in the 2026 Formula One Power Units [11]

Objective / Problem	Methodology / Technique	Dataset / Sample
Circuit-dependent deployment	Braking/straight energy calculations	2024 qualifying telemetry; Baku, Monza, Silverstone

Key Findings	Limitations / Research Gap	Relevance to Our Work
Shows scenario dependence across circuits	Student study; assumed flows; older telemetry	Context for cross-circuit scenarios

Study 12

S.No.	Author & Year	Paper Title
12	Jiashu Xu; 7 Nov 2023	Thermodynamic evaluation of 2026 Power Unit technical regulation changes in Formula 1 [12]

Objective / Problem	Methodology / Technique	Dataset / Sample
2026 PU regulation changes	Thermodynamic and regulation comparison	2023/2026 regulation comparison

Key Findings	Limitations / Research Gap	Relevance to Our Work
Discusses changed energy architecture	Conceptual analysis; not empirical SoC validation	Background; verify current rules separately

3 Dataset and Exploratory Analysis

3.1 Source and provenance

The telemetry source is TracingInsights/2026 [15]. The newer importer pins commit e8cf4fe5121447409fa0cb3728255f261cfc8d62 and preserves source JSON bytes, URLs and checksums. It selects event/Race/ANT lap telemetry, laptimes and session metadata. Original lap identifiers and original lap start times are retained. The historical Australian CSV is separately identified by SHA-256 9f12948c5da1f8386f4b6ba4c36c0795d82ddbc39a954b4879375b6095171b21; its results must not be silently relabeled as results from the newer import.

Table 1. Completed race-import coverage reported by the multi-track implementation and associated experiment artifacts.

Race	Samples	Laps	Driver	Analysis role
Australia	37,569	58	ANT	Temporal and cross-track ML; energy scenarios
Austria	39,416	71	ANT	Temporal and cross-track ML; energy scenarios
Italy	36,453	53	ANT	Temporal and cross-track ML; energy scenarios

These imports total 113,438 records and 182 race laps. Broader acquisition/preprocessing coverage remains separate from this evaluated subset. No complete season-wide session manifest was established for this report.

3.2 Data dictionary

Field	Meaning and use	Units / caveat
time / session_time	Chronology and integration intervals	Seconds; newer import uses original lap start offsets
lap_number / lap_time	Original lap identity and local timing	Available in newer import; historical laps inferred
speed	Road-load and ML input	km/h; convert to m/s
rpm; gear	Operating context and classifier input	rpm; integer gear
throttle; brake	State-label construction and flow gates	Percent throttle; binary brake in audited race CSV
distance; rel_distance	Lap-position selection and checks	Metres; fraction
acc_x/y/z	Source acceleration context	Units/processing not independently calibrated
x/y/z	Recorded trajectory display	Source coordinate units unverified
DistanceToDriverAhead	Traffic context	Source distance convention; not electrical input

No channel supplies battery current, terminal voltage, temperature, usable capacity, true SoC, measured motor power or a calibrated ICE map.

3.3 Preprocessing

The original fetch/preprocessing scripts and newer race adapter form different data stages. The original fetcher discarded lap IDs and assembled cumulative time from lap durations. Historical race laps are inferred when relative distance wraps from above 0.75 to below 0.25. The newer importer instead preserves original IDs and uses original lap starts plus lap-local time; it rejects missing lap files, noncontiguous IDs, mismatched arrays and reversed timestamps. This preserves long interruptions rather than compressing them.

Numerical conversion preserves invalid/missing values as null. Signals are checked for finite values, diagnostic ranges, chronology, duplicates and gaps. DriverAhead, when present, is categorical and its None value represents missing context. Diagnostic plausibility thresholds are flags, not proven physical limits. Strong braking is retained rather than deleted merely for its magnitude.

Speed is converted using v = speed/3.6. Acceleration is a causal least-squares slope over the preceding 0.6 s, requiring at least two valid points spanning more than 0.05 s. Fits do not cross gaps longer than 0.5 s. This derivative uses data through the current endpoint; it does not forecast future acceleration. Integration uses actual irregular intervals, not a fixed sampling frequency.

3.4 Exploratory findings

The historical Australian summary records 57 duplicate timestamps, five large gaps, zero invalid timestamps and 58 inferred laps. Equal-time boundary records contribute zero transferred energy. Recorded missing counts are zero for the listed numeric fields, but derivative availability and unsupported time gaps still affect modeling. The unknown modeled duration is 3.436 s.

Speed-integrated distance and recorded distance provide an independent numerical diagnostic, not a surveyed track calibration. For historical lap 2 the integrated distance is 5,148.002 m versus a recorded span of 5,146.269 m, a difference of 1.733 m. Other laps have larger discrepancies; they must not all be interpreted as a constant offset. Source-coordinate plots preserve row correspondence but cannot establish physical alignment to circuit graphics.

Figure 1 shows the historical race's conditional energy history and limited scenario spread. Figure 2 compares classification results. These plots describe actual saved outputs; scenario values are not observations of the car battery.

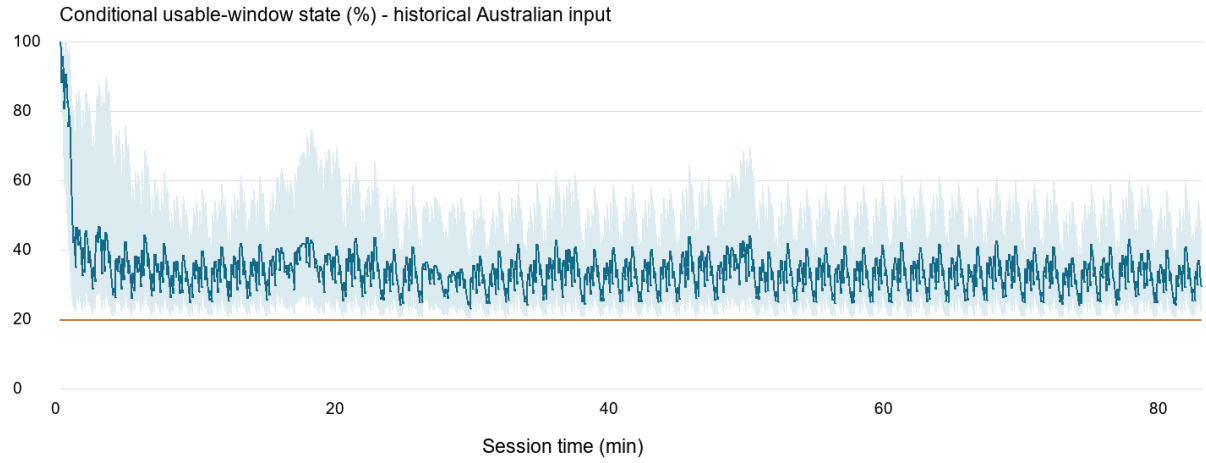


Figure 1. Historical Australian conditional usable-window state and scenario spread.

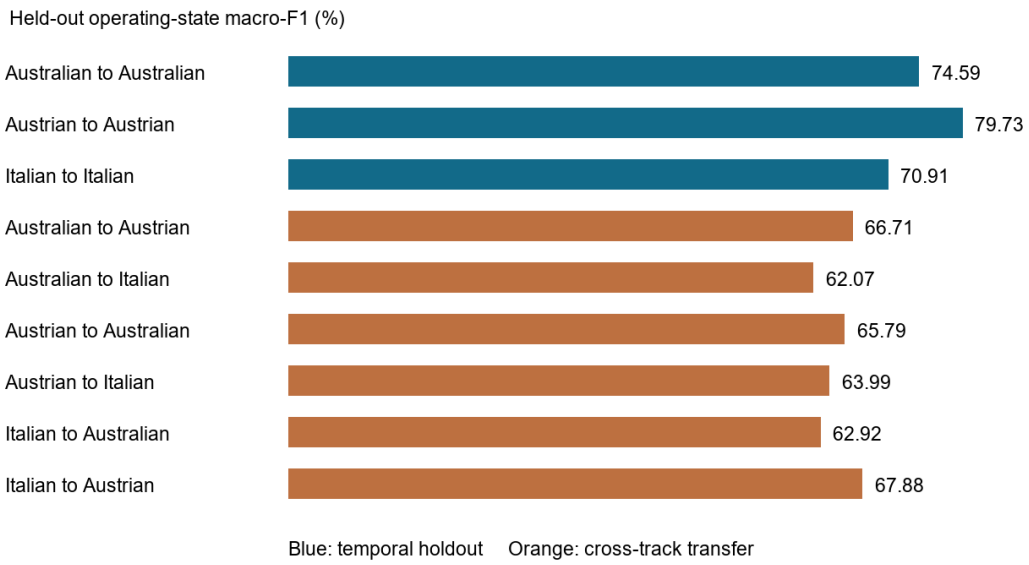


Figure 2. Temporal and cross-track operating-state classification macro-F1.

4 Mathematical Formulation

4.1 Longitudinal power demand

For speed v (m/s), acceleration a (m/s^2), mass m (kg), air density rho (kg/m^3), effective drag area CdA (m^2), rolling coefficient Crr and g = 9.80665 m/s^2:

F_req = m a + 0.5 rho CdA v^2 + Crr m g. (1)

P_w = F_req v / 1000 [kW]. (2)

Positive P_w represents propulsion demand; negative P_w represents net braking demand after the modeled road loads. Constant mass and effective aero are assumed. Grade, wind, fuel-burn variation, drivetrain losses and lateral tyre dynamics are omitted. These omissions can be mistaken for ERS contribution if the split is treated as observed.

4.2 Identifiability

An approximate propulsion relation is P_w ≈ P_ICE + P_K. For any admissible perturbation delta, P_ICE + delta and P_K - delta produce the same sum. Wheel demand therefore does not uniquely identify either component. Likewise, braking demand does not separate friction braking from regeneration. Integrating net electrical flow also leaves an initial-state ambiguity. Bounding these variables reduces the candidate set but does not prove a unique battery trajectory. ML operating-state labels do not supply the missing electrical measurements.

4.3 Conditional flow policy

Let d and r be nonnegative motor-side mechanical-equivalent deployment and recovery powers (kW). Let alpha be the assumed electric share and beta the recovery fraction. A propulsion opportunity is min(P_d,max, max(P_w,0)) when brake is off and throttle is positive; otherwise it is zero. Requested recovery is min(P_r,max, beta max(-P_w,0)) during braking and zero otherwise. The brake flag is a gate, not a calibrated regeneration sensor.

The reserve is E_res = q E_window. The deployment taper is h(E) = min(1, max(0,E-E_res)/(b E_window)), where q = 0.20 and b = 0.40 by default. Requested deployment equals alpha h(E) times the propulsion opportunity. A zero taper width disables gradual taper while retaining the reserve floor. Accepted deployment is limited by spendable energy; accepted recovery is limited by headroom. Charge and discharge cannot occur simultaneously under these gates.

4.4 Continuous energy accounting

For delta_t = t_k - t_(k-1), with powers attached to the ending sample:

E_k = E_(k-1) - d_k delta_t/(1000 eta_d) + eta_r r_k delta_t/1000. (3)

Energy E is in MJ; eta_d and eta_r are effective conversion efficiencies. Initial energy E_0 = E_window at each independent race’s first lap/time zero is imposed. Energy does not reset when laps change. Only optional lap-budget counters reset. The displayed SOC* = 100 E/E_window is a normalized usable-window fraction, not physical electrochemical SoC.

Define D = sum(d delta_t/1000), R = sum(r delta_t/1000), and L = sum(((1/eta_d-1)d + (1-eta_r)r) delta_t/1000). The accounting identity is:

E = E_0 - D + R - L. (4)

For unsupported intervals the representative state is held and unknown duration is accumulated. This is a conditional handling rule, not evidence of zero electrical flow. Duplicate timestamps cause no transfer. Accepted flow totals exclude rejected deployment and recovery.

4.5 Parameters and uncertainty

Parameter	Default	Interpretation
E_window	4 MJ	Modeled usable swing; not total battery capacity
m; rho	800 kg; 1.2 kg/m^3	Illustrative constant assumptions
CdA; Crr	1 m^2; 0.015	Uncalibrated effective road-load parameters
eta_d; eta_r	0.95; 0.80	Effective conversion assumptions
alpha; beta	0.30; 0.35	Assumed deployment share and recovery fraction
Deployment/recovery caps	350/350 kW	Research caps at model interface
q; b	0.20; 0.40	Reserve and taper fractions

Parameter	Default	Interpretation
Maximum supported gap	0.5 s	Missing-flow handling threshold
Lap energy budgets	null	Unenforced; not a claim of regulatory permission

Three scenarios use (alpha,beta) = (0.30,0.35), (0.15,0.50), (0.50,0.15). Their pointwise spread is a limited sensitivity illustration, not a statistical confidence interval or exhaustive feasible envelope. None establishes parameter identifiability. A wider calibrated sensitivity study over mass, drag, efficiencies and initialization remains future work.

4.6 Regulatory interpretation

FIA Section C Issue 16, published/approved 27 February 2026, specifies an electrical DC power ceiling and an energy-state swing in Articles C5.2.7 and C5.2.9, respectively [13]. Other clauses add speed/mode, harvesting and start restrictions. The implemented motor-side scenario caps do not directly certify these DC-interface requirements. The official index lists Issue 21, dated/published 19 October 2026 [14]; this later listing does not make that issue retroactively applicable to every earlier race. Event-specific rule mapping, mode information and electrical-interface conversion have not been established. Accordingly, all reported feasibility is model feasibility, not FIA compliance.

5 System Architecture and Implementation

The data path is: pinned source files to validated race adapter to recorded and calculated channels to conditional energy model to API to synchronized dashboard. Separate experiment paths take validated telemetry into the classifier and selected-lap baseline into the optimizer. Saved artifacts retain input/configuration identity.

Module	Responsibility
dataPreprocess/fetchKimi.py; preprocess.py	Historical acquisition and quality-control pipeline
scripts/import_races.py; backend/races.py	Pinned multi-race import and dataset registry
backend/data.py	Normalization, chronology and derived acceleration
backend/model.py	Conditional flows, continuous state and energy ledger
backend/optimizer.py	Constrained allocation and gradient diagnostics
backend/ml_validation.py; ml_experiments.py	Classification, baseline and isolated experiment results
backend/api.py	FastAPI telemetry, energy, selection and experiment endpoints
frontend/src	React/TypeScript dashboard with Plotly and recorded trajectory

The frontend shares one selected recorded sample across distance controls, chart cursors, readouts and map marker. Nearest-record lookup avoids inventing samples. The selected race determines lap options and energy context. Static circuit references are displayed separately where available; image annotations are not used as verified activation zones. Processed reference warnings are preserved rather than silently corrected into assumed truth.

The API distinguishes running, completed and failed jobs. JSON exports retain input hashes, parameters, source fingerprints and optimizer history. The current background-job design assumes one process; a durable queue would be required for distributed execution. A deployed production environment was not established by the supplied verification record.

6 Machine Learning Model Development

6.1 Task and labels

The implemented HistGradientBoostingClassifier recognizes contemporaneous operating states: braking when brake=1; otherwise coasting for throttle <=5%; partial throttle for 5%<throttle<95%; and high throttle for throttle >=95%. These are operational rules, not independently annotated semantic or electrical ground truth. Features are

current speed, RPM, gear and causal acceleration. Throttle and brake, which define the target, are excluded, along with lap, distance and estimated ERS outputs. The task is current-state recognition rather than advance prediction.

6.2 Training protocol

The fixed classifier uses 100 boosting iterations, 15 leaf nodes, learning rate 0.1, seed 42 and disabled early stopping. Within-track experiments train on the first floor(0.7L) whole laps, purge one entire lap and test on the remainder. Australia uses laps 1-40/41/42-58; Austria 1-49/50/51-71; Italy 1-37/38/39-53. Cross-track experiments train on every eligible source-race sample and test on eligible samples of a distinct target race. No target-track fitting or test-set hyperparameter tuning is documented.

A most-frequent training-class baseline is evaluated on the same test observations. Accuracy, balanced accuracy, macro precision/recall/F1, weighted F1, per-class support and confusion matrices are saved. For class c, precision=TP_c/(TP_c+FP_c), recall=TP_c/(TP_c+FN_c), and F1_c=2PR/(P+R). Macro-F1 averages classwise F1 equally. Undefined metrics use zero and retain class support. Since samples are autocorrelated, sample-independent confidence intervals are not asserted. Metric definitions follow the scikit-learn documentation [16].

7 Deployment Optimization

The selected-lap optimization begins at the baseline energy immediately before the first lap interval. It retains accepted baseline recovery and limits delivered energy to the baseline budget. Recorded speed is fixed, so the optimization cannot predict the speed response to a changed strategy.

Partition the lap into B=40 sequential bins. Decision x_j scales baseline deployment within bin j. Matrix A contains baseline delivered MJ by sample/bin, so Ax is the candidate delivered-energy vector. The baseline x=1 is representable; zero-baseline samples remain unavailable for new deployment. Let w=A^T(v/max(v)), let D be the adjacent-bin first-difference matrix and lambda=0.02. The objective is:

maximize J(x) = w^T x - lambda ||Dx||^2. (5)

Equivalently minimize f=-J, with analytic gradient:

grad f = -w + 2 lambda D^T D x. (6)

Constraints enforce nonnegative multipliers, samplewise opportunity caps, available energy, storage headroom, the configured reserve and an energy budget no greater than baseline deployment. Recovery is fixed. Each energy state is an affine function of x; therefore this is a convex quadratic objective with linear constraints under the stated fixed inputs. SciPy SLSQP [17] uses analytic derivatives, a default 150-iteration limit and 10^-8 objective tolerance. Final feasibility is checked at 10^-6 tolerance. This is constrained gradient-based numerical optimization, not a separately implemented plain gradient-descent training algorithm.

The proxy favors energy at higher recorded speed and penalizes abrupt multiplier changes. It is disclosed as an allocation preference, not an established approximation to minimum lap time. Downstream laps are not counterfactually recomputed. A terminal reserve and full-race control study would require a different horizon and performance model.

8 Results and Performance Evaluation

8.1 Evidence versions

Historical documentation records a pre-reserve state trajectory ending at zero modeled energy. The newer saved reproduction uses reserve_fraction=0.20 and taper_fraction=0.40. These are different policies. The numerical energy and optimizer results below use the reserve-aware saved artifacts, while the nine ML experiments use the separately fingerprinted newer race imports. Mixing them as a single experimental run would be misleading.

8.2 Conditional energy results

Quantity	Saved reserve-aware Australian result
Samples / inferred laps	37,569 / 58

Quantity	Saved reserve-aware Australian result
Initial SOC*	100% (imposed)
Final SOC*	29.483590%
Maximum conservation residual	8.55316e-13 MJ
Unsupported duration	3.436 s
Independent battery-state error	Undefined: no electrical reference

A small conservation residual confirms arithmetic consistency of accepted model flows. It does not establish that the assumed flows occurred in the car. The reserve changes depletion behavior by construction. Consequently, a higher final modeled energy than in older exports is not an experimentally demonstrated estimator improvement.

8.3 Classification results

Table 2. Saved experiment metrics; percentages are rounded to two decimals. Baseline means the most frequent source-training class.

Mode	Train to Test	Train n / Test n	Accuracy %	Macro-F1 %	Baseline accuracy %	Baseline macro-F1 %
within track	Australian to Australian	26259 / 10570	87.98	74.59	54.40	17.62
within track	Austrian to Austrian	27152 / 11468	90.72	79.73	50.33	16.74
within track	Italian to Italian	26207 / 9368	90.65	70.91	70.21	20.62
cross track	Australian to Austrian	37462 / 39156	85.91	66.71	48.91	16.42
cross track	Australian to Italian	37462 / 36211	82.76	62.07	59.96	18.74
cross track	Austrian to Australian	39156 / 37462	81.92	65.79	49.02	16.45
cross track	Austrian to Italian	39156 / 36211	83.17	63.99	59.96	18.74
cross track	Italian to Australian	36211 / 37462	80.04	62.92	49.02	16.45
cross track	Italian to Austrian	36211 / 39156	85.22	67.88	48.91	16.42

All nine classifiers exceed their most-frequent baseline on accuracy and macro-F1. Temporal results are generally stronger than transfer results, consistent with circuit-dependent operating conditions. Accuracy exceeds macro-F1 substantially, particularly in Italy, so majority-class performance must not substitute for balanced class analysis. Cross-track macro-F1 of 62.07-67.88% shows a meaningful remaining transfer gap. These comparisons are descriptive; no significance test or multi-driver external validation is claimed.

8.4 Optimizer result

Quantity	Baseline	Optimized
Proxy objective	1.31313006	1.52988419
Delivered energy (MJ)	1.93042427	1.93042427
Final energy (MJ)	1.39180430	1.39180430

Run 5621b1851aaadff3 concerns historical Australian lap 2 and converged in 79 iterations. Initial energy is 1.39138514 MJ; maximum saved constraint violation is 2.22e-16. The proxy increases by 16.51%. This is a proxy improvement at the same delivered-energy budget, not a measured race-time gain.

8.5 Software verification

The supplied verification record documents backend accounting/chronology tests, synthetic edge cases, optimizer analytic-versus-numerical gradient checks, API checks, shared frontend selection checks, lint and production build. Its earlier count of 15 backend and five frontend tests belongs to that historical verification stage; it is not asserted as a fresh result for the newer checkout. This report independently inspected saved result structures and reconciled numerical values; it does not present unexecuted full-system checks as new successful tests. Synthetic fixtures validate implementation behavior, not real battery accuracy.

9 Discussion

The principal outcome is a working conditional analysis framework. Its equations reveal how uncertainty in road load, electric share and recovery fraction affects energy. Explicit reserve logic produces plausible-looking trajectories, but appearance cannot identify hidden electrical behavior. Multiple ICE/motor splits can explain the same motion; a single scenario remains only one conditional interpretation.

The ML component adds a tested telemetry-recognition capability with a transparent baseline and leakage-aware partitions. Its usefulness is limited to the operational labels actually evaluated. It is not a learned residual correcting ERS energy, despite the earlier handbook's proposed hybrid estimator. No classifier output closes the energy observation gap.

The optimizer demonstrates derivative-based constrained allocation and consistent energy budgeting. Its support restriction, fixed speed and fixed recovery simplify the numerical problem but also remove the vehicle-response mechanism required to establish lap-time optimality. A larger proxy value should therefore be interpreted within that numerical objective only.

Transferable engineering principles include preserving chronology, separating measurements from assumptions, enforcing dynamic constraints and auditing cross-domain performance. Commercial HEV/PHEV or fleet applications would need vehicle-specific models, measured current/voltage, thermal limits and independent validation. No road-vehicle deployment benefit has been demonstrated.

10 Limitations

1. Electrical ground truth and true initial battery state are absent; battery-state MAE/RMSE cannot be computed.

2. The road-load and power-split parameters are illustrative, and the state is structurally non-identifiable.

3. Braking-only recovery omits possible non-braking harvesting; friction braking is unresolved.

4. State holds across unsupported gaps do not model physical evolution, particularly during long interruptions.

5. Scenario spread is not a calibrated uncertainty interval; broad physical sensitivity and ablations remain incomplete.

6. Completed multi-track evaluation covers three races and one driver, not every available 2026 session.

7. Mode signals, event-specific limits and electrical-interface conversion are insufficient for compliance certification.

8. The optimizer changes a fixed-speed proxy over one lap and cannot establish improved lap/race time.

9. Source acceleration and coordinates are uncalibrated; reference graphics cannot prove activation or exact track alignment.

10. The selected literature review has uneven full-text access. Detailed sample sizes or source performance figures unavailable for verification are not asserted.

11 Conclusion and Future Scope

The project integrates continuous conditional ERS energy accounting, telemetry-state classification and constrained deployment allocation in a reproducible software workflow. The reserve-aware Australian reproduction preserves numerical conservation, the three-track ML experiments outperform a simple baseline on their stated labels, and the selected-lap optimizer improves its declared proxy without increasing delivered energy. These are distinct results

with distinct validation limits.

Future work should first freeze a complete event/session manifest and event-applicable regulation snapshots, then obtain independent electrical reference data or an instrumented simulator. Calibrated road-load and powertrain models would support wider sensitivity tests and feasible-set reconstruction. ML residual estimation should be attempted only with defensible targets. A validated vehicle-response model, continuous counterfactual race propagation and robust terminal-energy constraints are prerequisites for lap-time or race-time optimization. Additional drivers, sessions and circuits are needed for broader ML generalization claims.

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Appendix A Reproducibility and Evidence Register

Evidence	Report use	Interpretation
Class_report/send2AI/tempt	Supplied archive snapshot	Earlier Australian-only implementation and broad preprocessing material
docs/MULTI_TRACK.md; backend/races.py	Newer multi-race workflow	Distinct from archive snapshot
results/model_summary.json; reproduction.json	Reserve-aware state/accounting	Historical Australian input; imposed policy
results/optimization/5621b1851aaadff3.json	Lap-2 optimizer figures	Reserve-aware run; not the older 56-iteration result
results/ml_experiments/*.json	Nine classification experiments	Fingerprinted newer imports; true ERS labels absent
config/model.json	Parameter defaults	Assumptions, not measured car properties
docs/VALIDATION.md	Historical software verification	Earlier completed checks, not a new report-time test run
Literature DOCX and XLSX	Candidate bibliography	Source summaries require access-aware verification

For reproduction, provision the pinned source data, install the locked dependencies, run scripts.import_races and scripts.prepare_experiments, and start scripts.dev. Historical energy reproduction uses scripts.reproduce --lap 2. These commands write into the application's configured data/results locations; they are documented rather than rerun during report authoring. Report-specific scripts, figures and evidence extracts are retained inside Class_report.

Appendix B Metric and Class Details

The rows below preserve per-class held-out F1 and support for each temporal experiment. A zero or low minority-class score must remain visible rather than being hidden by overall accuracy.

Australian Grand Prix

Class	Precision	Recall	F1	Test support
braking	0.8234	0.8521	0.8375	1663
coasting	0.6116	0.2580	0.3629	531

Class	Precision	Recall	F1	Test support
partial_throttle	0.8574	0.8264	0.8416	2626
high_throttle	0.9150	0.9697	0.9416	5750

Austrian Grand Prix

Class	Precision	Recall	F1	Test support
braking	0.8607	0.9325	0.8952	2267
coasting	0.6120	0.3735	0.4639	490
partial_throttle	0.8983	0.8442	0.8704	2939
high_throttle	0.9454	0.9747	0.9598	5772

Italian Grand Prix

Class	Precision	Recall	F1	Test support
braking	0.7692	0.9136	0.8352	1065
coasting	0.4158	0.2049	0.2745	205
partial_throttle	0.8802	0.6765	0.7651	1521
high_throttle	0.9437	0.9804	0.9617	6577

Appendix C Assumption and Completion Register

Item	Status	Required treatment
Initial SOC* = 100%	Imposed	Top of modeled usable window, not measured battery charge
Continuous cross-lap accounting	Implemented	Independent races initialize separately
Physics/ML residual hybrid	Proposed, not completed	Do not describe classifier as ERS residual learning
Three scenario trajectories	Implemented	Limited scenario sensitivity, not full uncertainty calibration
All available 2026 sessions	Intended scope	Full acquisition/evaluation not established
Gradient-based selected-lap proxy optimization	Implemented and saved	No established lap-time benefit
Absolute SoC validation	Unavailable	Requires independent battery observations
FIA event compliance	Unverified	Requires issue/event/interface/mode mapping

Appendix D AI Tool Usage Report

An AI assistant assisted with organizing supplied materials, reconciling implementation stages, locating bibliographic and regulatory sources, drafting text and equations, and generating report tables and figures from saved project artifacts. AI-generated prose is not experimental evidence. Numerical project claims are attributed to identified data/configuration/result files; inaccessible literature details are not represented as independently reproduced findings. The authors must review technical statements, citations, attribution and institutional disclosure requirements before submission. This disclosure describes report preparation; it does not assert that the assistant authored the underlying project implementation.